Project Design Phase-II
Data Flow Diagram & User Stories

Date	27 October 2026
Team ID	LMP-EV-2026
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

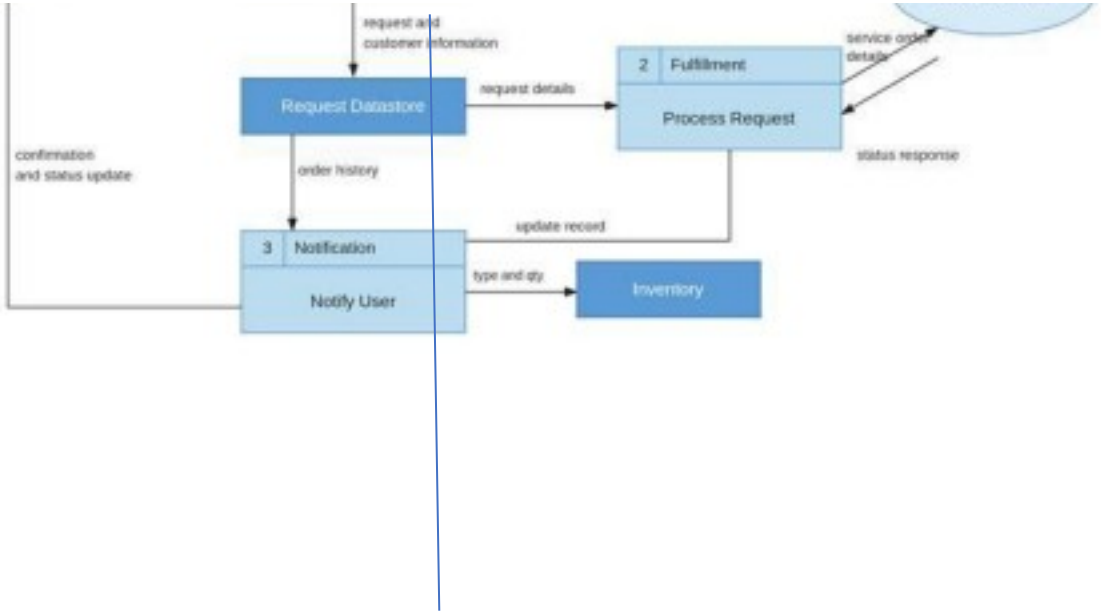
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

ample: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Dashboard Viewer / EV Strategy Director	Dashboard access	US-01	As an EV market strategy director, I want to open the public EV dashboard from the project website so that I can explore the published analysis.	The GitHub Pages site opens without sign-in; the embedded Tableau view loads or provides a working direct Tableau Public link.	High	Release 1
Dashboard Viewer / EV Strategy Director	Executive summary	US-02	As a dashboard viewer, I want to see annual sales, average price in USD, average range and average customer rating in one summary so that I can understand headline indicators.	The Executive Summary displays the four KPI values with labels and units; the average price is identified in USD ($).	High	Release 1
Dashboard Viewer / EV Strategy Director	Category analysis	US-03	As a dashboard viewer, I want to compare annual sales across vehicle categories so that I can identify higher- and lower-sales categories.	The Category Sales worksheet ranks categories by annual sales units and updates when applicable filters change.	High	Release 1
Dashboard Viewer / EV Strategy Director	Manufacturer analysis	US-04	As a dashboard viewer, I want to compare manufacturer sales contribution in a treemap so that I can examine relative manufacturer scale.	The Manufacturer Share treemap represents sales units by manufacturer and shows labels and values.	High	Release 1
Dashboard Viewer / EV Strategy Director	Battery technology	US-05	As a dashboard viewer, I want to compare battery technology share and median driving range so that I can examine adoption mix and range differences.	Battery Tech Share displays battery-type percentages; Battery Performance displays median range for LFP, NCA, NMC, Sodium-Ion and Solid-State.	High	Release 1
Dashboard Viewer / EV Strategy Director	Country and battery analysis	US-06	As a dashboard viewer, I want to compare vehicle counts by manufacturing country and battery type so that I can inspect their distribution.	Country vs Battery Matrix shows battery type by country with vehicle counts and a readable legend.	Medium	Release 1
Dashboard Viewer / EV Strategy Director	Geographic analysis	US-07	As a dashboard viewer, I want to view annual sales by manufacturing country on a map so that I can explore the geographic footprint in the dataset.	The Global Manufacturing Map displays country-level sales values and responds to applicable dashboard filters.	Medium	Release 1
Dashboard Viewer / EV Strategy Director	Product positioning	US-08	As a dashboard viewer, I want to compare average price in USD and average range by vehicle category so that I can explore product-positioning trade-offs.	Price vs Range displays category marks with average range and average price in USD axes; viewers can inspect marks and tooltips.	High	Release 1
Dashboard Viewer / EV Strategy Director	Customer and technical metrics	US-09	As a dashboard viewer, I want to compare average customer rating and average battery capacity across categories so that I can review the measures together.	Rating vs Battery shows customer-rating bars and the battery-capacity line on the displayed dual-axis view.	Medium	Release 1
Dashboard Viewer / EV Strategy Director	Interactive filtering	US-10	As a dashboard viewer, I want to filter by model year, category, battery type and manufacturing country so that I can focus on a relevant market subset.	The specified filters can be applied and applicable dashboard worksheets respond consistently.	High	Release 1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Dashboard Viewer / EV Strategy Director	Story and recommendations	US-11	As a dashboard viewer, I want to open the completed Story Point 1, Global EV Market & OEM Overview, so that I can review the category sales and manufacturer contribution narrative.	Story Point 1 opens and displays its intended Category Sales and Manufacturer Share (Treemap) visuals. Story Points 2-4 are planned and are not yet available.	Medium	Release 1 - Story Point 1 only
Project Owner (Yuvraj Shrivastava)	Workbook and website maintenance	US-12	As the project owner, I want to update the source CSV connection, republish Tableau and verify the website embed so that the public view reflects the approved data version.	The updated workbook is published to Tableau Public and the website embed or direct link opens the current view. Refresh is manual, not scheduled.	High	Manual update

PROJECT-SPECIFIC DATA FLOW DIAGRAMS

System boundary: EV market CSV source, Tableau Desktop authoring, Tableau Public hosting, and a static HTML/CSS website on GitHub Pages. The website embeds the Tableau Public view. No separate application database, API backend, Flask service, Streamlit service or scheduled ETL process is evidenced.

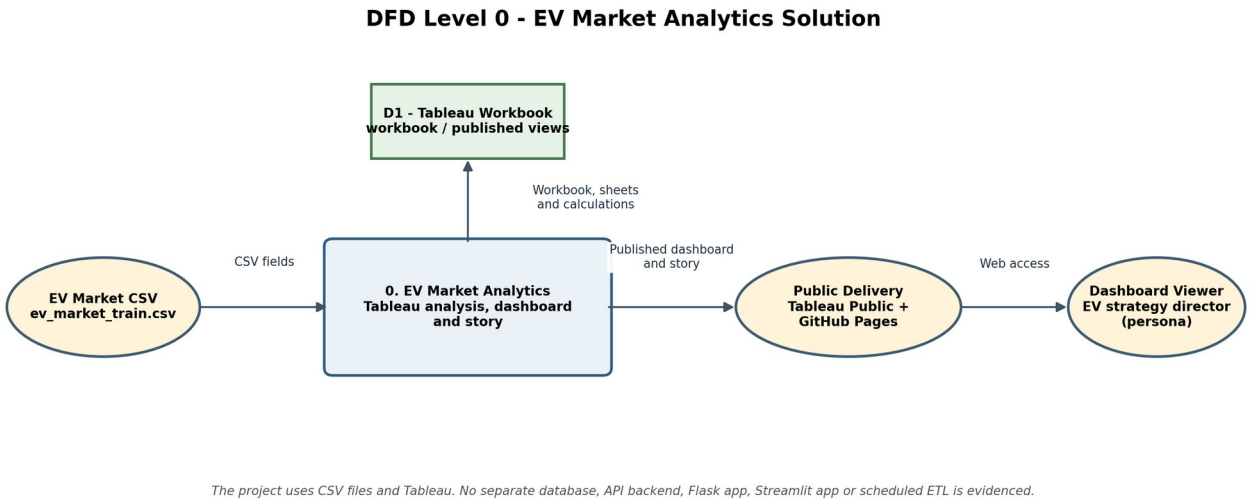


Figure 1. DFD Level 0 context diagram for the EV Market Analytics solution.

DFD LEVEL 1 - INTERNAL DATA FLOW

The ev_market_train.csv source is connected in Tableau Desktop. During preparation, null values and duplicate records are removed and data types are corrected; fields are then configured and aggregated to build the executive KPIs and nine worksheets. The workbook is published to Tableau Public and embedded in the GitHub Pages website. Source updates are performed manually by the sole project owner.

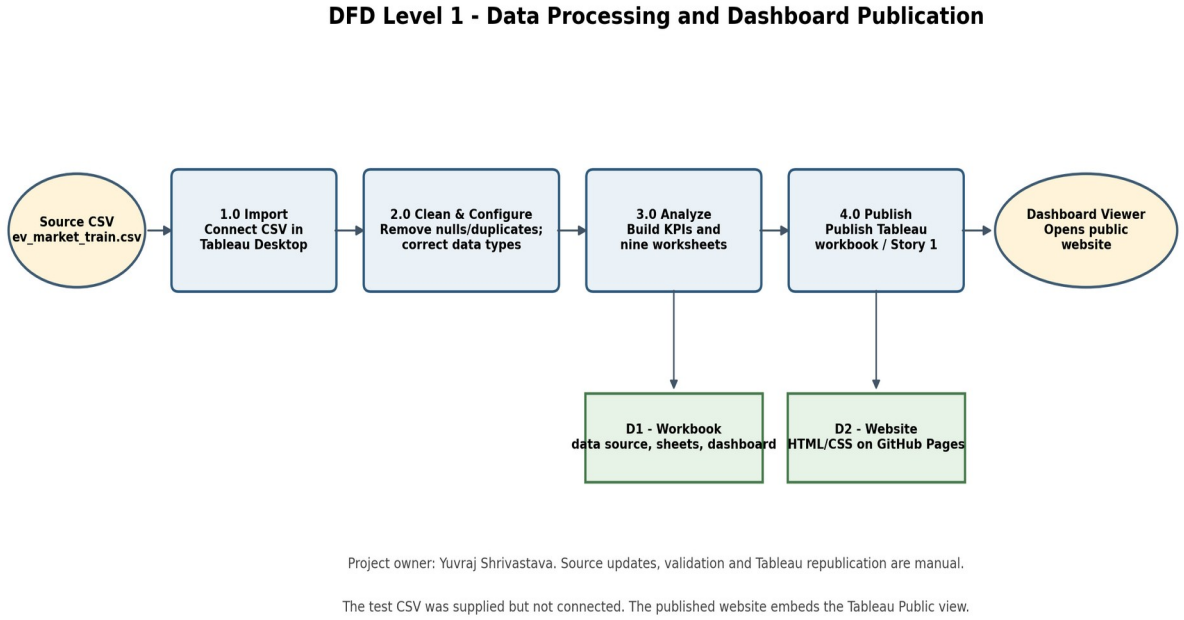


Figure 2. DFD Level 1 showing import, field configuration, analysis, publishing and web delivery.

Key data sources and outputs

- Source: ev_market_train.csv (shown in the supplied Excel screenshot and connected to the Tableau workbook). Null values and duplicate records were removed and data types were corrected during preparation. ev_market_test.csv was supplied as a resource but was not connected to Tableau. The dataset schema includes vehicle, manufacturer, model year, manufacturing country, region, category, battery specifications, range, efficiency and market measures.
- Outputs: Executive Summary; Category Sales; Manufacturer Share (Treemap); Battery Performance; Battery Tech Share; Global Manufacturing Map; Price vs Range; Rating vs Battery; Country vs Battery Matrix.
- Publication: Tableau Public dashboard and completed Story Point 1 (Global EV Market & OEM Overview), embedded on the GitHub Pages website. Story Points 2-4 are planned and are not completed. Sole project owner: Yuvraj Shrivastava (EN25CS3011105), Medicaps University.